

INARI VAYLA, Finland

WMO: 028270

Lat: 69.0722N

Lon: 27.4925E

Elev: 124

StdP: 99.85

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-30.9	-37.3	0.1	-33.5	-34.3	0.2	-30.8	11.5	-5.1	10.2	-5.3	1.4	320	0.762

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	24.1	16.9	21.7	15.6	19.6	14.6	17.7	22.5	16.6	20.7	15.3	18.8	4.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.0	11.5	19.4	14.7	10.6	18.1	13.7	9.9	17.0	50.5	22.5	47.0	20.7	43.3	18.9	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
9.3	8.1	7.1	-37.8	26.7	2.4	2.3	-39.6	28.4	-41.0	29.7	-42.4	30.9	-44.1	32.6		
			-37.2	19.0	2.7	2.1	-39.2	20.5	-40.8	21.7	-42.3	22.9	-44.2	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.6	-13.9	-13.1	-8.5	-1.7	4.9	9.7	13.4	10.9	7.1	0.0	-6.5	-10.2
	DBStd	10.83	7.90	8.53	6.12	4.58	4.03	3.98	3.92	3.45	3.20	4.27	5.87	7.07
	HDD10.0	4101	740	647	573	351	167	54	11	29	97	311	494	627
	HDD18.3	6914	998	880	831	601	416	262	158	232	337	569	744	885
	CDD10.0	237	0	0	0	0	9	43	117	57	10	0	0	0
	CDD18.3	10	0	0	0	0	0	2	7	2	0	0	0	0
	CDH23.3	116	0	0	0	0	5	13	82	16	0	0	0	0
	CDH26.7	25	0	0	0	0	1	2	19	4	0	0	0	0
	WSAvg	3.2	2.9	2.9	3.7	3.3	3.4	3.6	3.2	3.0	3.2	3.2	3.0	3.1
	PrecAvg	449	27	24	22	24	31	53	68	59	40	42	30	30
Precipitation	PrecMax	594	50	62	41	64	72	108	137	164	92	100	63	53
	PrecMin	277	13	5	10	6	9	4	23	23	8	8	9	16
	PrecStd	73	9	12	9	12	16	26	32	29	17	23	13	10
	PrecStd	73	9	12	9	12	16	26	32	29	17	23	13	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	3.2	4.7	10.4	22.7	25.1	29.3	24.9	17.9	10.2	4.5	3.3
		MCWB	0.8	1.6	1.9	6.2	14.0	16.5	18.6	18.0	13.9	8.2	3.4	1.8
	2%	DB	-0.2	1.2	2.7	7.5	18.3	21.9	25.6	21.7	15.8	8.5	2.6	1.6
		MCWB	-1.0	0.1	0.8	3.7	11.0	14.2	17.4	16.4	12.3	7.2	1.8	0.6
	5%	DB	-2.1	-0.2	1.5	5.7	14.7	19.3	22.7	19.2	14.2	7.0	1.3	0.3
		MCWB	-2.7	-1.0	-0.1	2.5	9.4	13.2	16.2	15.0	11.4	5.9	0.8	-0.4
	10%	DB	-3.7	-1.7	0.3	4.3	11.9	16.6	20.6	17.3	12.3	5.5	0.5	-0.8
		MCWB	-4.2	-2.4	-1.0	1.7	7.6	12.0	15.7	14.1	10.4	4.8	0.1	-1.4
	0.4%	WB	1.0	2.0	2.2	6.4	14.2	17.6	19.7	19.1	14.4	8.4	3.5	2.1
		MCDB	2.3	2.7	3.9	10.6	21.8	23.2	23.7	23.8	16.6	9.6	4.1	3.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-1.0	0.3	1.2	4.0	11.5	15.9	18.4	17.1	13.0	7.4	1.9	0.8
		MCDB	-0.3	1.1	2.6	7.0	16.8	20.6	23.8	20.8	15.4	8.5	2.5	1.5
	5%	WB	-2.6	-1.1	0.1	2.9	9.6	14.0	17.2	15.6	11.8	6.1	0.9	-0.2
		MCDB	-2.0	-0.2	1.2	5.0	14.3	17.3	21.8	18.5	13.7	6.9	1.2	0.3
	10%	WB	-4.2	-2.4	-0.9	2.1	8.0	12.5	16.1	14.4	10.7	4.8	0.2	-1.4
		MCDB	-3.7	-1.7	0.3	3.9	11.5	16.1	20.0	16.9	12.3	5.5	0.5	-0.8
Mean Daily Temperature Range	5% DB	MDBR	10.0	9.6	11.3	9.6	9.0	8.7	9.2	9.1	7.6	5.3	6.8	9.0
		MCDBR	9.2	6.3	7.5	9.3	14.6	12.8	13.5	12.5	10.2	5.5	4.8	5.8
		MCWBR	9.0	5.6	6.2	6.2	7.9	6.3	6.5	7.4	6.9	4.4	4.6	5.6
		MCDBR	9.0	6.7	6.8	8.3	13.0	11.7	12.1	11.2	9.2	5.1	4.6	5.8
	5% WB	MCWBR	8.7	6.1	5.9	5.7	7.4	6.4	6.4	7.0	6.5	4.3	4.5	5.6
		taub	0.196	0.247	0.290	0.322	0.344	0.343	0.355	0.337	0.305	0.264	N/A	N/A
		taud	1.913	2.058	2.145	2.217	2.393	2.476	2.464	2.492	2.546	2.261	N/A	N/A
		Ebn at Noon	39	503	719	812	843	856	829	804	740	495	N/A	0
All-Sky Solar Radiation	RadAvg	0.02	0.42	1.88	3.72	4.62	4.78	4.41	3.10	1.63	0.52	0.07	0.00	0.00
	RadStd	0.00	0.04	0.11	0.22	0.49	0.43	0.49	0.29	0.15	0.07	0.01	0.01	0.00

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page